

Muhammad Abdullah

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EDUCATION AND TRAINING

02/11/2022 – 10/08/2026 Lahore, Pakistan
BACHELOR OF SOFTWARE ENGINEERING Lahore Garrison University, DHA, Lahore

Principal subjects: Machine Learning; Big Data Analytics; Data Structure and Algorithms; Linear Algebra; Calculus and Analytical Geometry; Probability and Statistics; Stochastic Processes, Operational Research, Discrete Structures, Theory of Automata, Programming Fundamentals, Object-Oriented Programming, Database Systems, Software Design and Architecture, Software Construction and Development, and Final Year Project in Medical Imaging Computer Vision. AI

Level in EQF EQF level 6

WORK EXPERIENCE

AI/ML ENGINEER (FULL-TIME, JUN 2026–PRESENT) | JR. AI/ML ENGINEER (PART-TIME, SEP 2025–MAY 2026) – AILYTICA – 01/09/2025 – Current – LAHORE, PAKISTAN

Medical Vision & Feature Segmentation: Designed and deployed hybrid CNN-Transformer models (Swin-UNet, MedSAM adaptations) for automated 2D/3D feature extraction and multi-class image segmentation, achieving sub-second GPU latency and high Dice similarity scores across target datasets.

MLOps, Quantization & Inference Infrastructure: Engineered end-to-end MLOps pipelines utilizing vLLM and TensorRT-LLM to serve quantized models (AWQ/QAT), reducing VRAM consumption by 45% while doubling inference throughput across Docker-containerized microservices.

Edge Vision & Real-Time Tracking: Built real-time visual tracking and spatial feature-extraction microservices using OpenCV, TensorRT, and PyTorch, optimizing processing speed to 30+ FPS for deployment on resource-constrained embedded nodes.

GraphRAG & Knowledge Retrieval: Developed enterprise GraphRAG architectures integrating Graph Neural Networks (GNNs) with vector databases (Qdrant), improving retrieval contextual accuracy by 35% on complex structured datasets.

Dataset Engineering & Model Benchmarking: Built automated data preprocessing, synthetic augmentation, and benchmarking scripts in PyTorch for deep learning backbones (UNet, ResNet, EfficientNet), establishing baseline metrics for downstream production tasks.

ACADEMIC & AI LAB RESEARCH PROJECTS

01/01/2026 – 22/05/2026
RetinaXplain - Hybrid CNN-Vision Transformer Framework for Diabetic Retinopathy Classification

Final Year Project-Sole AI Researcher and Developer

Data Pipeline & Preprocessing: Processed 3,662 APTOS images across 5 DR classes using circular cropping and Ben Graham normalization, expanding data to ~16,000 augmented instances.

Model Training & Algorithmic Features: Trained and evaluated Swin-T (85.27% accuracy, 0.7216 macro-F1), EfficientNet-B3, and ViT-B/16, engineering weighted soft-voting ensembles, clinical label remap,& agreement analysis.

App Deployment & Documentation: Built a multi-page web app featuring real-time DR diagnostic reports, probability visualizers, session analytics, and downloadable JSON records; authored a 72-page technical report.

Links <https://github.com/abdullahse2211/diabetic-retinopathy-classification> | <https://youtu.be/MnS1vS5ZLqU?si=td3M1nW6vMmZRVf5>

07/01/2026 – 08/03/2026
BioNAFNet - Low-Light Image Enhancement | Student Deep Learning Researcher

Novel Architecture & Loss Formulation: Designed a bio-inspired enhancement network trained from scratch with a composite Charbonnier, SSIM, color, and Weber-Fechner perceptual loss, surpassing 9 baselines with 28.67 dB PSNR on LOLv2 Real.

Knowledge Distillation & Acceleration: Compressed model parameters from 154M to 24M via knowledge distillation (6.4x compression, 3.9x speedup) with zero perceptual degradation, enabling the student model to surpass the teacher by +0.29 dB.

01/12/2025 – 02/02/2026

ThalCare - Alpha Thalassemia Classification | Student ML Researcher

Clinical Feature Engineering: Engineered 15 diagnostic features combining 9 CBC/HPLC parameters with 6 clinical indices (Mentzer, Shine & Lal, Green & King), elevating Silent Carrier diagnostic F1 to 0.92–0.955.

Robust Multi-Class ML Benchmarking: Evaluated 6 ML architectures (hybrid CNN-Transformer, 1D-CNN, SVM, RF, LR) for 5-class differential diagnosis, achieving 90.60% balanced accuracy and 0.906 weighted F1 under injected clinical noise and 10% feature dropout.

SKILLS

AI, Vision & Frameworks:

PyTorch, TensorFlow, OpenCV, scikit-learn, MedSAM, Swin-Transformer, UNet3+, Diffusion, RAG

Systems, MLOps & GenAI:

VLLM, TensorRT, Triton, Docker, 4-bit AWQ, Qdrant, GraphRAG, Qwen2-VL, LlamaIndex, LoRA, BoTorch

Languages & Tools:

Python, C++, SQL (MySQL), MATLAB, NumPy, Pandas, Matplotlib, Streamlit, Git, GitHub, Overleaf

CERTIFICATIONS

Certiport, a Pearson VUE Business, 13/01/2025

Information Technology Specialist: JavaScript

International industry certification in JavaScript. Credential ID: mTp4-DwVM

Mode of learning: Presential

Link <https://verify.certiport.com/>

National Vocational & Technical Training Commission (NAVTTTC), 16/12/2024

JavaScript Full-Stack Development (MERN)

Completed full-stack JavaScript development training covering the MERN technology stack: MongoDB, Express.js, React, and Node.js. Grade A+.

Mode of learning: Project based

University of Engineering and Technology, Lahore National Centre of Artificial Intelligence, 08/12/2023

Prompt Engineering - Certificate of Completion

Completed practical training in prompt design and refinement for generative AI

Mode of learning: Presential

IBM; Coursera, 18/11/2023

IBM Web Development

Completed IBM training in foundational web-development concepts, tools, and practices.

Mode of learning: Online

Link <https://coursera.org/verify/E8X42H6HASJL>

LANGUAGE SKILLS

Mother tongue(s): **URDU**

Other language(s): **ENGLISH**